Explainable artificial intelligence

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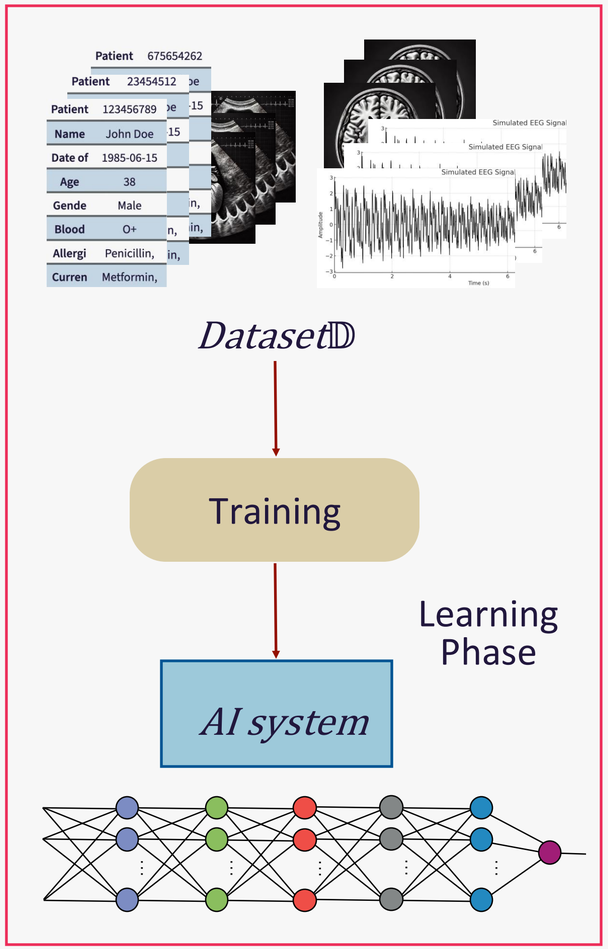
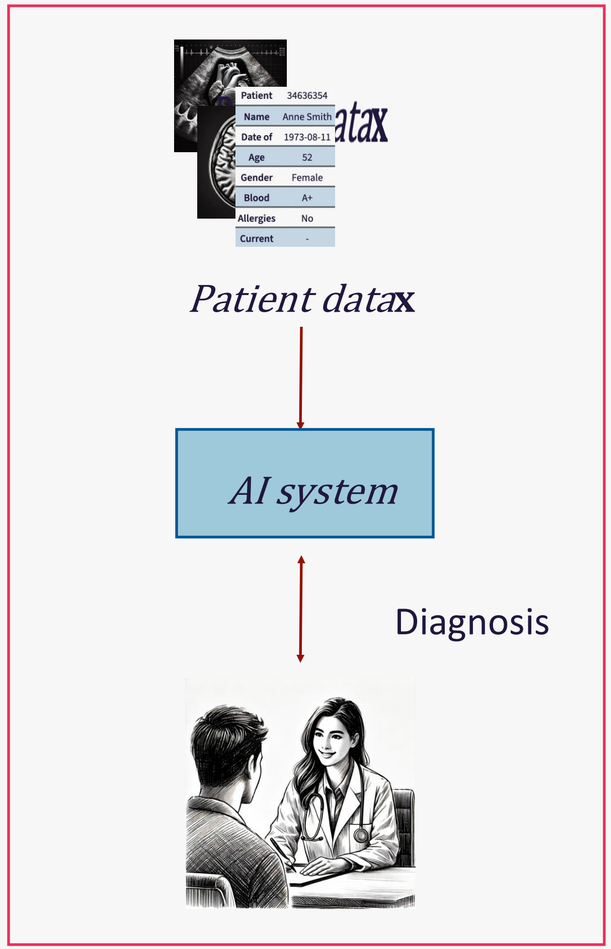
Providing meaningful explanations

- Data: What kind of data was the system trained on?
- Output: What kind of output does the system give?
- Performance: How accurate/precise/reliable are the predictions?
- How: How does the system make predictions?



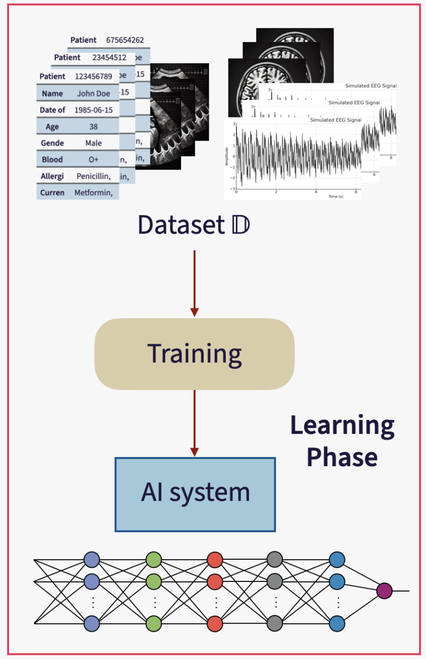


synthetic data, created by chatGPT; sketch created by DALL·E

- Why: Why/how is this instance given this prediction?
- Why not: Why is this instance NOT predicted to be [a different outcome Q]?
- How to be that: How should this instance change to get a different prediction Q?
- How to still be this: What is the scope of change permitted for this instance to still get the same prediction?
- What if: What would the system predict if this instance changes to…?

XAI Question Bank
Q. Vera Liao, Daniel M. Gruen, Sarah Miller: Questioning the AI: Informing Design Practices for Explainable AI User Experiences. CHI 2020: 1-15

Explanations in AI-assisted medical diagnosis

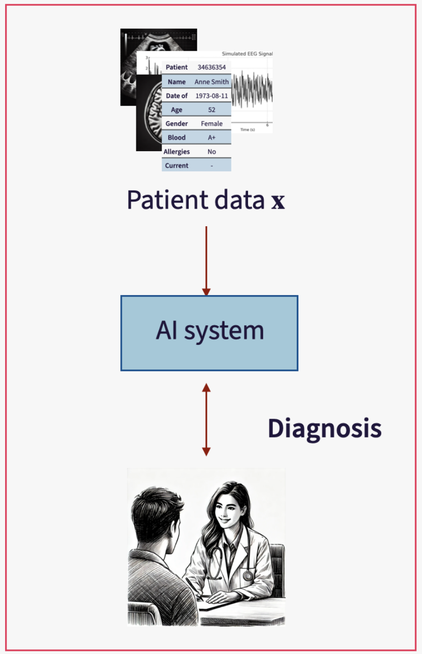


These examples of AI explanations are entirely fictional, created using our imagination, knowledge retrieved from the web and generative AI (ChatGPT)

Data: What kind of data was the system trained on?

- Source. “I was trained on a dataset consisting of chest X-ray images from multiple hospital networks, including publicly available sources such as the NIH ChestX-ray14 dataset, MIMIC-CXR, and private datasets collected from partner hospitals. These datasets include de-identified medical images obtained from radiology departments and curated for research purposes.”
- Labeling. “I was trained on a dataset consisting of radiology images labeled by board-certified radiologists. Each image was annotated with diagnostic labels such as ‘Pneumonia,’ ‘Tuberculosis,’ and ‘No Abnormality’ based on radiology reports and clinical findings. In some cases, consensus labeling was applied, meaning multiple experts reviewed and agreed upon the diagnosis to improve label reliability.”
- Size. “I was trained on a dataset consisting of 1.2 million medical images, including 500,000 CT scans, 400,000 chest X-rays, and 300,000 MRI scans. The dataset was divided into 80% for training, 10% for validation, and 10% for testing to ensure model generalization.”
- Bias. “I was trained on a dataset consisting of chest X-rays primarily from patients in North American hospitals, which may introduce a geographic and demographic bias. This means my performance might be less accurate for patients from underrepresented populations, such as those from low-resource settings or with rare diseases that are less frequent in the training data.”

Explanations in AI-assisted medical diagnosis



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Why/how is this instance given this prediction? How should this instance change to get a different prediction?

- Feature analysis. "My prediction is that this patient has pneumonia, because the chest X-ray shows areas of increased opacity in the lower left lung region, which is consistent with lung consolidation. Additionally, the shape and distribution of these opacities match typical radiographic patterns of pneumonia seen in my training data."

- Comparative analysis (instance-based). "My prediction is that Instance A is classified as a normal heart rhythm, while Instance B is classified as atrial fibrillation, because Instance B's ECG shows irregular RR intervals and absence of distinct P waves, which are strong indicators of atrial fibrillation. In contrast, Instance A exhibits regular sinus rhythm with clearly defined P waves, leading to different predictions."

- Comparative analysis (label-based). "My prediction is that this brain MRI scan indicates a benign tumor (P) rather than malignant cancer (Q), because the detected lesion is well-defined, non-invasive, and does not exhibit irregular borders or rapid growth characteristics typically seen in malignant tumors. The probability score for malignancy was below my decision threshold, so I classified it as benign."

- Counterfactual. "My prediction is that this patient does not have pneumonia, but if the opacity in the lower lung field were larger and more consolidated, I would predict pneumonia instead. Additionally, an increase in C-reactive protein (CRP) levels in the clinical data could also shift my prediction from 'No Pneumonia' to 'Pneumonia'."